Distributed System Overview - Assignment

Assignment

You are implementing **4 specific pieces** of a distributed key-value store in Java, to make consistent hashing, replication, and the CAP theorem concrete instead of theoretical. Everything else (storage engine, node plumbing, cluster orchestration, the CP-mode reference implementation) is **already written** for you; you're not building the whole system from scratch.

1. Unpack and Import

- Unzip kv-store-template.zip.
- Open the kv-store-template folder in your IDE as a Maven project (accept the auto-import prompt).
- Wait for dependencies (JUnit 5) to resolve.

2. Verify It Builds

- cd kv-store-template
- mvn compile - Expected: BUILD SUCCESS.

Problem Statement

Only these 4 TODOs are required:

#	Class	What to implement
1	ring/ConsistentHashingStrategy	Hash ring + virtual nodes
2	node/ClusterNode	3 one-line delegations to store
3	replication/ReplicationManager	SYNCHRONOUS branch only
4	consistency/APConsistencyStrategy	Fallback-to-replica logic

Also remove annotation - `@Disabled("Disabled until trainee completes the required TODO files")` from test files and make sure all test cases pass successfully.

Details -

The tasks are:

- Implement consistent hashing in ConsistentHashingStrategy so keys are assigned to nodes using a ring and virtual nodes.
- Implement storage delegation in ClusterNode so put/get/delete operations use the underlying KeyValueStore.
- Implement synchronous replication in ReplicationManager so writes to followers complete before returning.
- Implement AP read/write behavior in APConsistencyStrategy so reads fall back to a replica when the preferred node is unreachable and writes do not throw on degraded availability.

The integration test suite is currently disabled until those TODOs are completed

Deliverables:

- **Code submission** - Submit your completed implementation as a zip file (or via the submission method specified separately) containing the full project with your changes.
- **Scope of changes** - Your submission should only modify:
    - ring/ConsistentHashingStrategy
    - node/ClusterNode
    - replication/ReplicationManager
    - consistency/APConsistencyStrategy
    - Removal of the @Disabled(...) annotations on the integration test files
- **Proof of passing tests** - Include a screenshot (or pasted terminal output) of mvn test showing all tests passing - specifically the integration test suite that was previously disabled.

[End]